Maël Rabot

Software Engineer · Developer Mentoring & Robotics

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PROFILE

Software engineer completing a Master's in Computer Science at Epitech. I mentor developers at Epitech International and co-founded Lucy-Robotics, where we build Lucy, an open-source framework that makes robots simpler to use: one platform, any robot. I enjoy low-level programming, object-oriented design, web development and robotics, and I like working in international, team-driven environments.

EXPERIENCE

Pedagogical Mentor · Epitech International

Paris, France (on-site) · 2026 – Present · Fixed-term contract

- Mentor 150+ international exchange students from 25 countries, alongside MSc 1 & 2 students, through project-based learning in English
- Run technical code reviews and project defenses across C, C++, Python, TypeScript/Vue.js, Elixir and DevOps
- Apply Socratic mentoring to build autonomy, architectural thinking and debugging skills rather than handing over solutions
- Evaluate software architecture, code quality, testing and design patterns; follow each student on organization, work habits and well-being
- Coordinate the program behind the scenes: activity schedules and liaison with Erasmus managers, French teachers and administration

Co-Founder & Product Manager · Lucy-Robotics

Lyon, France · Oct. 2024 – Present · Non-profit association

- Co-founded the association behind Lucy, an open-source robotics framework: one ROS 2 core between robots and the interfaces that drive them
- Lead the product roadmap and engineering teams: backlog prioritization, sprint planning and agile delivery with Kanban
- Contribute to the ROS 2 framework architecture and its technical documentation
- Drive open-source hardware integration and 3D-printed prototyping

Full Stack Developer · AudioWizard

Lyon / Remote · July 2023 – August 2025 · Contract

- Built and maintained SaaS products for hearing care professionals and their clinical workflows (TypeScript, Node.js, React)
- Administered PostgreSQL databases and optimized query performance
- Deployed a ForestAdmin back-office for internal operations
- Owned continuous integration and production maintenance (Docker)

Research & Development Developer · PoC Innovation Lab

Lyon, France · January 2023 – August 2023 · Student research association

- Researched and prototyped deep learning models that turn images into musical scores (Python, PyTorch)
- Implemented convolutional neural networks and generative AI architectures; managed experimental datasets and data pipelines

Workshop Lead & Mentor · Coding Club

Lyon, France · February 2023 – February 2025 · Association

- Organized and hosted hands-on coding workshops in Python, C and web development for high-school students
- Mentored students through competitive programming and hackathon challenges; designed activities with educators

SKILLS

Languages
C, C++, Python, TypeScript, JavaScript, Rust, Go, Elixir, Haskell, SQL, Bash

Web
React, Vue.js, Node.js, Vite

Robotics
ROS 2, ros2_control, URDF, InMoov, 3D printing

Data, AI & tools
PostgreSQL, MariaDB, PyTorch, Git, Docker, CI, Agile

Engineering
Software architecture, code review, OOP, ECS, networking

Soft skills
Mentoring, teamwork, autonomy, problem solving

EDUCATION

Master's in Computer Science
Epitech Lyon
2022 – 2027
C/C++, Unix system programming, network architecture · 3.2 GPA · School ambassador · Innovation Hub contributor

International Engineering Program
Chung-Ang University, Seoul
2025 – 2026
Exchange year: distributed systems, robotics

Scientific Baccalaureate
Lycée International de Ferney-Voltaire
2018 – 2021

LANGUAGES

French: native
English: C1 (professional)

SELECTED PROJECTS

Lucy: open-source robotics framework · Lucy-Robotics

`Active · ROS 2, ros2_control, C++, Rust, Python, TypeScript, React · lucy-robotics.com`

> One platform // Any robot. A common layer between robot hardware and the tools that drive it, so the same software works whatever the robot.

- Core: all the ROS 2 logic in C++, Rust and Python; main packages lucy_ros_packages (built on ros2_control) and lucy_embedded_firmware
- lucy_ws: one workspace with Python scripts and installers, running in a few commands on macOS, Windows and Linux without Docker (pixi + RoboStack)
- Robots supported today: InMoov (open-source 3D-printed humanoid by Gaël Langevin) and the SO-101 arm; robot-agnostic by design
- Interfaces: Lucy Control Panel (React / TypeScript web app for real-time control and telemetry), a CLI, and any custom interface
- Open-source ecosystem: GPL-3.0 software, CC BY-NC hardware, public documentation, roadmap, issue tracking and Discord community

HuRI: human-robot interaction · Lucy-Robotics

`Active · Python, ROS 2 · github.com/Lucy-Robotics/HuRI`

- AI framework for speech-to-speech conversation, multi-layer memory (short, medium, long term) and kinematic grounding, so robots converse and move naturally

R-Type: networked game engine · ASM Studios

`C++, CMake, ECS · github.com/ASM-Studios/R-Type`

- Cross-platform ECS game engine recreating the arcade classic R-Type, with networked multiplayer

AREA: automation platform · ASM Studios

`TypeScript, Vite, Go, Docker, OAuth2, 2FA · github.com/ASM-Studios/AREA`

- Recreation of the IFTTT action-reaction automation platform, with OAuth2 and two-factor authentication

Not A Rhythm Game

`Active · C++, ImGui, GLFW, BASS · github.com/Mael-RABOT/NotARhythmGame`

- Open-source rhythm game inspired by Muse Dash, with a complete level editor and a custom audio engine

Corewar · ASM Studios

`C, assembly, compiler design · github.com/ASM-Studios/ASM-corewar`

- Virtual machine arena and bytecode assembler where competing programs fight for survival in memory

Pandoc Parser

`Haskell, functional programming · github.com/Mael-RABOT/Pandoc`

- Document format parser and converter written in Haskell, built on AST transformations

Teams TUI Clone

`C, BSD sockets, networking · github.com/Mael-RABOT/microsoftTeams`

- Lightweight terminal clone of Microsoft Teams with multi-client messaging over network sockets

AI Number Recognition

`Python, machine learning, computer vision · github.com/Mael-RABOT/AI_number_recognition`

- Neural network built from scratch to recognize and classify hand-drawn digits

OTHER EXPERIENCE

Delegate · Model United Nations (FerMUN)

`Ferney-Voltaire / Geneva (UN) · September 2019 – June 2021`

- Represented delegations in debates held at the ITU and the UN in Geneva on international technology and sustainability policy
- Public speaking and diplomatic negotiation in English and French; drafted multilateral resolutions and position papers

INTERESTS

Robotics & 3D printing · Open source · Mentoring · Aviation · Game development & graphics programming